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RESEARCH NOTE

Tectonic rotation of the Santa Ynez Range, California, recorded in the Sespe Formation

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SUMMARY

Terrestrial claystone and siltstone in the Sespe Formation (Oligocene) in the Santa Ynez Range near Santa Barbara, California, record a palaeomagnetic field that predates the regional folding and satisfies a reversal test. The mean declination of 80.1° closely matches the declination in Miocene volcanic and sedimentary rocks studied by others in the Western Transverse Block (of which the Santa Ynez Range is a part), and the mean inclination of 38.9° is consistent with a hypothesis that the block was once about 10° south of its present latitude. The combined palaeomagnetic data indicate that clockwise rotation of the block is entirely post-Oligocene; the rotation might not have commenced until the middle Miocene, or about 15 Myr after northward motion of the block began.

Key words: Oligocene, palaeomagnetism, Sespe Formation, southern California tectonics.

INTRODUCTION

A quarter of a century ago the possibility was raised that the Transverse Ranges in the Western Transverse Block—a region north of Los Angeles bracketed by the Santa Ynez, San Gabriel, and Dume faults (Luyendyk *et al.* 1985; Fig. 1)—are rotated clockwise relative to the structural grain of southern California (Hamilton & Myers 1966). More recently it was shown palaeomagnetically that the block is indeed rotated clockwise by about 80° (Kamerling & Luyendyk 1979; Luyendyk, Kamerling & Terres 1980; Hornafius 1985; Kamerling & Luyendyk 1985; Luyendyk *et al.* 1985; Terres & Luyendyk 1985; Hornafius *et al.* 1986; Liddicoat 1988). It was also recognized that since the middle Miocene the rotation has been episodic (Hornafius *et al.* 1986), and sedimentary rocks of Pleistocene age are rotated by as much as 30° (Levi *et al.* 1983). On the basis of an inclination anomaly in the Miocene rocks, it was hypothesized that the block was at a more southerly latitude (west of Baja California) when the rotation commenced (Crouch 1979; Kamerling & Luyendyk 1979). Although not fully understood, the structural and kinematic implications of those findings warrant careful examination (Sylvester 1988).

The palaeomagnetic investigations cited above were made on Miocene volcanic rocks in the western Transverse Ranges, Santa Monica Mountains, and northern Channel Islands, and on the sedimentary Monterey Formation (Miocene) and Saugus Formation (Pliocene–Pleistocene) near Santa Barbara and Ventura, respectively. Attempts were also made to use Eocene and older submarine-fan deposits on San Miguel Island (Champion, Howell & Marshall 1986) and Eocene sandstone on Santa Rosa Island (Kamerling & Luyendyk 1985) to establish the magnitude of the rotation in the northern Channel Islands. In this research note palaeomagnetic data are reported for the Oligocene Sespe Formation in the Santa Ynez Range that further document the tectonic history of the Western Transverse Block.

LOCALITIES, STRATIGRAPHY, AND AGE

The Sespe Formation was sampled at two localities in Santa Barbara County (Fig. 2). The westernmost locality is in a road cut about 200 m south of the vehicle stop on the north lane of US 101 at Gaviota Pass (34.5°N, 239.8°E; Sites 1–4, Table 1). The formation there is transitional between marine and non-marine, consisting of interbedded gray siltstone and red claystone that dip 60° to the south. The other locality (34.5°N, 240.2°E; Sites 5–8, Table 1) is also a road cut. It is about 5 km north of Santa Barbara along California Highway 154 where non-marine gray sandstone and siltstone are interbedded with red claystone. At both

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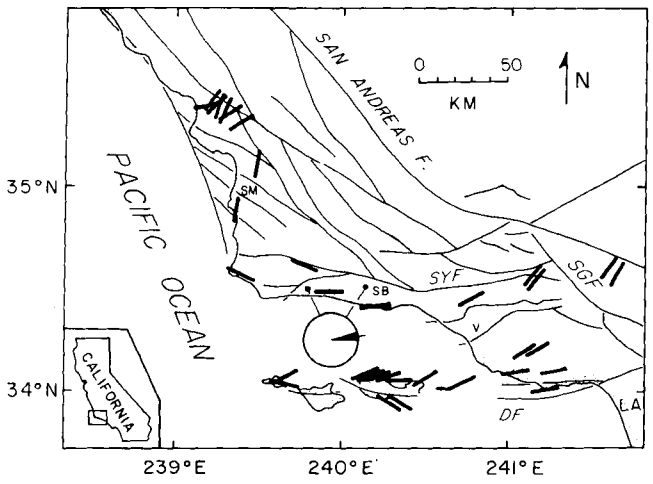


Figure 1. Present fault pattern in southern California from Jennings (1973) and Dibblee (1976). The Western Transverse Block of Luyendyk *et al.* (1985) is defined by the Santa Ynez Fault (SYF) on the north, the San Gabriel Fault (SGF) on the east, and the Dume Fault (DF) on the south. The block has been rotated clockwise about 80° since the Oligocene as indicated by the palaeomagnetic declination (bold solid lines). Circle with the mean declination (80.1° ± 6.9°) is for the Sespe Formation in Santa Barbara County. Modified after Hornafius (1985) and Liddicoat (1988).

localities, the Sespe Formation is flanked by the Vaqueros Sandstone and Alegria or Gaviota formations (Dibblee 1950, 1966).

The red colour of the claystone and absence of marine fossils or marine-type sediments are evidence that the Sespe Formation has a terrestrial origin. As interpreted by Dibblee (1950, 1966), who mapped the region, the setting was a nearly level plain that developed after an Eocene sea became filled with sediments eroded from the east. A total of 700–1400 m of stream-laid sediment accumulated during gradual sinking of the area that fell behind the influx of sediment. The bordering sea retreated westward in the vicinity of Gaviota Pass where subareal deposits grade into marine littoral sand of the Alegria Formation. The iron oxide that colours the claystone and siltstone was derived from iron-bearing minerals of the source rocks. The weathering and oxidation reflect a climatic condition of generally high temperature and alternating wet and dry seasons (Dibblee 1966).

Although no marine fossils have been found within the Sespe Formation, some water-worn shells of *Ostrea idriaensis* were recovered near the base of the formation. However, the Oligocene age of the formation can be inferred by its stratigraphic position in a section of known marine deposits. For instance, the Sespe Formation is above the Coldwater Sandstone (Upper Eocene) and below the Vaqueros Sandstone (Lower Miocene). Moreover, the Gaviota and Alegria formations that, as mentioned, underlie the Sespe Formation contain Refugian molluscan and foraminiferal faunas (Dibblee 1966). On the basis of those faunas, the present author has assigned an age of 30 Myr to the Sespe Formation where sampled for this study, recognizing that this age is approximate and could be in error by several million years or more.

SAMPLING AND LABORATORY PROCEDURES

At both localities red-coloured claystone or siltstone was sampled, and for all but one site (where there are five specimens) six or more specimens were taken (Table 1). During a reconnaissance study at Gaviota Pass the

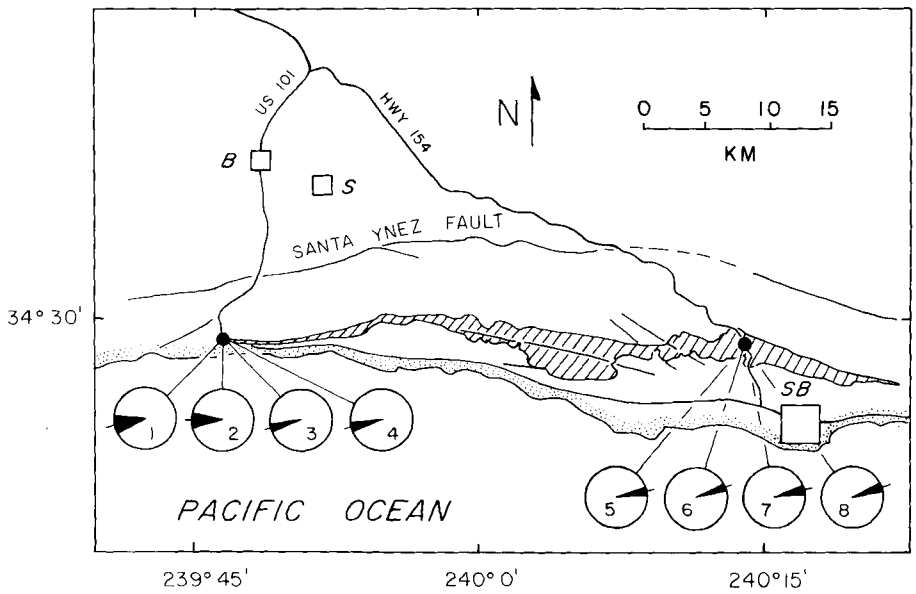


Figure 2. Location map of palaeomagnetic sites in the Sespe Formation (hachures) between Gaviota Pass (Sites 1–4) and Santa Barbara (Sites 5–8), Santa Barbara County, California. Sites 1–4 record reverse palaeomagnetic polarity and Sites 5–8 record normal palaeomagnetic polarity. Shaded region in each circle is the mean declination and ΔD; $\Delta D = \arcsin(\sin\alpha_{95}/\cos I)$. Note that the declination has an approximate east–west orientation; the mean declination for the combined sites is 80.1° ± 6.9° (Table 1) when the declination for Sites 1–4 is inverted through the origin. Exposure of the Sespe Formation is from Dibblee (1950).

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Table 1. Palaeomagnetic data for the Sespe Formation (Oligocene), Santa Barbara County, California.

SITE	I1	D1	I2	D2	N	k	α_{95}	ΔI2	ΔD2	F	ΔF	PLAT	PLONG
1	-17.1	231.4	-41.6	266.4	5	36.6	12.8	12.8	17.1	12.4	10.3	-16.1	131.5
2	-18.6	235.6	-38.5	270.6	6	49.1	9.7	9.7	12.6	15.5	8.0	-11.6	131.4
3	-15.3	227.3	-44.3	262.0	6	187.3	4.9	4.9	7.2	9.7	4.5	-20.6	131.7
4	-15.0	230.1	-33.0	262.2	6	42.6	9.4	9.4	11.0	21.0	7.7	-16.3	138.9
5	37.1	61.1	32.7	78.6	6	104.0	6.6	6.6	7.9	21.3	5.7	19.2	321.5
6	44.5	48.6	44.1	73.5	7	172.3	5.1	5.1	7.2	9.9	4.6	27.2	315.8
7	46.8	50.3	45.1	76.8	6	102.2	6.7	6.7	9.5	8.9	5.7	25.0	314.0
8	31.8	55.7	30.4	71.0	6	78.7	7.6	7.6	8.8	23.6	6.4	24.6	326.8
1-4	16.5	51.1	39.4	85.3	4	196.0	6.6	6.6	8.6	14.6	5.7	16.1	314.0
5-8	40.2	54.2	38.1	74.9	4	100.6	9.2	9.2	11.7	15.9	7.6	24.0	320.0
1-8	28.3	52.5	38.9	80.1	8	170.0	5.4	5.4	6.9	15.1	4.8	20.1	316.9

SITE: Site number; 1-4 is mean for sites of reverse polarity inverted through the origin, 5-8 is mean for sites of normal polarity, 1-8 is mean for all sites using normal polarity
I1: Mean inclination (degrees) prior to bedding correction and following thermal demagnetization at 400°C
D1: Mean declination (degrees) prior to bedding correction and following thermal demagnetization at 400°C
I2: Mean inclination (degrees) following bedding correction and thermal demagnetization at 400°C
D2: Mean declination (degrees) following bedding correction and thermal demagnetization at 400°C
N: Number of specimens (15-cc cubes) measured
k: Fisher (1953) precision parameter (N-1/N-R) for I2, D2
α_{95}: Alpha-95: radius (degrees) of circle of 95-percent confidence about the mean palaeomagnetic direction for I2, D2
ΔI2: α_{95}
ΔD2: $\sin^{-1}(\sin \alpha_{95}/\cos I2)$
F: Flattening of inclination (I2)
ΔF: Adjusted uncertainty from Beck (1980) using conversion factor 0.78 from Demarest (1983); Oligocene reference pole is 86.0°N, 154.0°E from Irving and Irving (1982)
PLAT: Latitude (+ north, - south) of Virtual Geomagnetic Pole (VGP) for I2, D2
PLONG: Longitude (east) of VGP for I2, D2

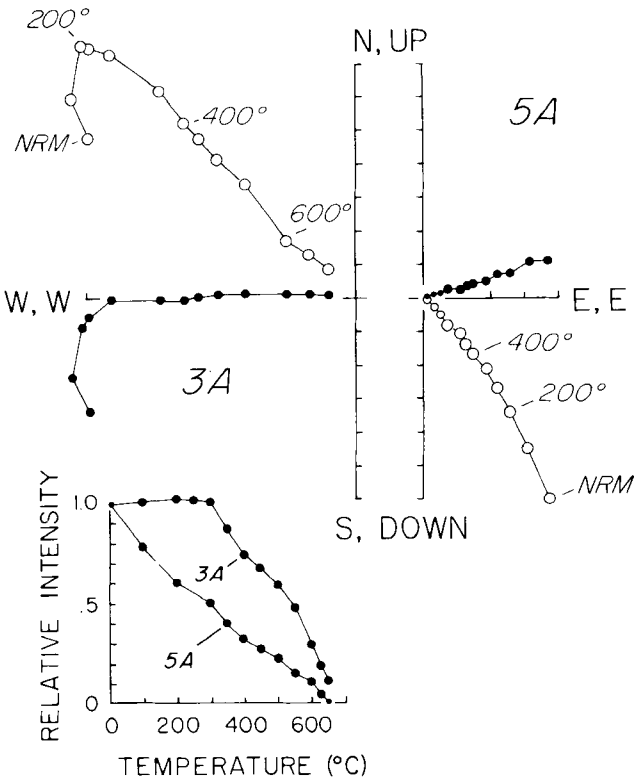


Figure 3. Vector diagram and normalized intensity plot for specimens 3A (reverse polarity) and 5A (normal polarity) from Sites 3 and 5, respectively, in the Sespe Formation, Santa Barbara County, California. The samples have a blocking temperature above 650 °C. In the vector plots, solid circles are projections on the NS–EW plane and open circles are projections on the EW–vertical plane; divisions on the axes are 1.00×10^{-3} A m^{-1}.

interbedded gray siltstone was also sampled, but subsequently found to have been remagnetized (failed a bedding test). At Gaviota Pass the sites are about 5 m apart stratigraphically, and along Highway 154 they are evenly spaced over several tens of metres of section.

The reconnaissance specimens were progressively demagnetized in alternating fields up to 99.9 mT or to 650 °C in Schonstedt equipment, and the remanence was measured in a cryogenic magnetometer. Although alternating field demagnetization reveals the correct palaeomagnetic polarity, the component that pre-dates the regional folding is isolated most accurately by thermal demagnetization above 400 °C (Fig. 3), the temperature used to demagnetize the remaining specimens. The maximum blocking temperature of the characteristic direction of magnetization is about 650 °C, identifying haematite as the source of the remanence.

PALAEOMAGNETIC DATA AND DISCUSSION

The Sespe Formation records reverse palaeomagnetic polarity at Gaviota Pass (Sites 1–4) and normal polarity along California Highway 154 (Sites 5–8, Fig. 4). The combined mean directions for each polarity (Table 1) give a positive reversal test at the 95 per cent level of confidence, and the directions in stratigraphic coordinates (corrected for bedding) indicate the characteristic direction of magnetization was acquired prior to the most recent folding in the Pliocene and Pleistocene (Dibblee 1966; Fig. 5). The grouping of directions is quite good—except for Site 1 where α_{95} is 12.8°, the sites have α_{95} values of under 10°. For each site the mean declination is deflected very much to the east,

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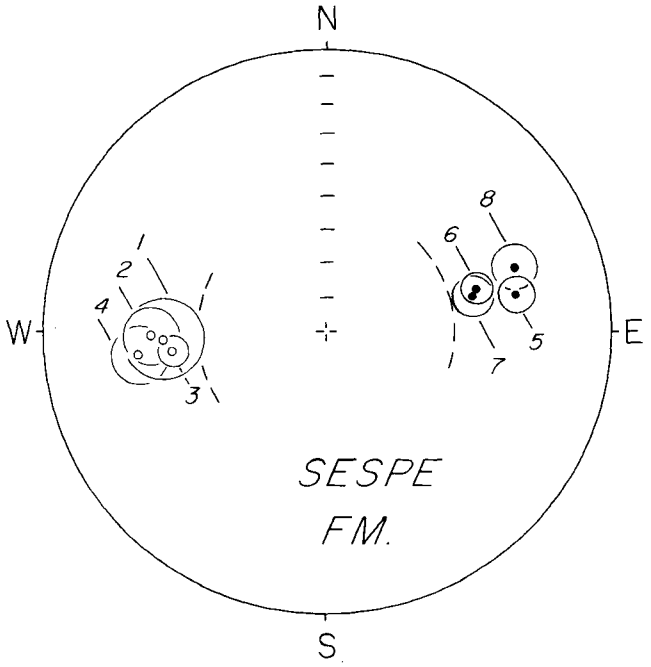


Figure 4. Equal-area plot of the mean palaeomagnetic directions and circles of 95 per cent confidence for Sites 1–8 in the Sespe Formation following thermal demagnetization at 400 °C (Table 1). Note that the mean inclination for each site is shallow when compared to the inclination of the expected Oligocene field (dashed circle; Irving & Irving 1982). Solid circles are plotted on the lower hemisphere, open circles on the upper hemisphere.

and inclination is shallow (flat) when compared to the inclination of the expected Oligocene field as predicted by Irving & Irving (1982). Inverting the directions for Sites 1–4 and combining the data for the eight sites produces a mean declination of 80.1° ± 6.9° and flattening of 15.1° ± 4.8°. The data thus confirm the large clockwise retation of the Miocene volcanic rocks in the Santa Monica Mountains (Kamerling & Luyendyk 1979) and Monterey Formation (Hornafius 1985) near Santa Barbara, and support the possibility of a more southerly latitude for the Western Transverse Block in the Oligocene.

The possibility of a southerly position was developed by Kamerling & Luyendyk (1979) by invoking the hypothesis of Crouch (1979) that major lithologic belts that comprise the California continental borderland and western Transverse Ranges were originally west of Baja California. Northward displacement is linked to a change in relative plate motions from subduction to right-lateral strike–slip motion along the southern California–Baja California continental margin (Crouch 1979).

If northward motion of the Western Transverse Block did not occur as the palaeomagnetic data would suggest, then the reasons presented by Kamerling & Luyendyk (1985) for an inclination anomaly might apply. Possibilities include: (1) insufficient averaging of secular variation, (2) incorrect measurement of the bedding attitude, especially in volcanic rocks, and (3) inclination error (King 1955) in sediment. Regarding the Sespe Formation, it would appear that enough time is spanned (at least one polarity boundary—several thousand years) by the two localities to average secular variation, even though the angular standard deviation of the Virtual Geomagnetic Poles for the combined sites is low (7.0°, +3.4°, −1.7°, $N = 8$) when compared to an expected angular standard deviation of about 12° (Doell & Cox 1972). As to the correct measurement of strike and dip in the field, the bedding is well developed in the Sespe Formation where the samples were taken, and the measured attitudes correspond to those on the geologic map of Santa Barbara County (Dibblee 1950, 1966); thus, the nearly 15° of flattening cannot be explained by field orientation alone. Neither is the inclination flattening in the Sespe Formation attributed to a depositional inclination error because the remanence most likely is a chemical remanent magnetization (Irving 1964) following growth of the iron oxide that imparts the red colour to the fine-grained beds.

Overall, the present east–west alignment of the Western Transverse Block reflects the geologic history of southern California since the Jurassic (Jones, Blake & Rangin 1976). However, as summarized by Hornafius (1985), opinions differ about the origin of the major tectonic elements. Among the possibilities are *in situ* development of the structural features by vertical tectonic processes (Reed & Hollister 1936; Dibblee 1950, 1966; Baird *et al.* 1974), positioning of the province by strike–slip faulting and counterclockwise rotation along the North American–Pacific plate boundary (Hall 1978, 1981a, b; Dickinson 1979, 1982,

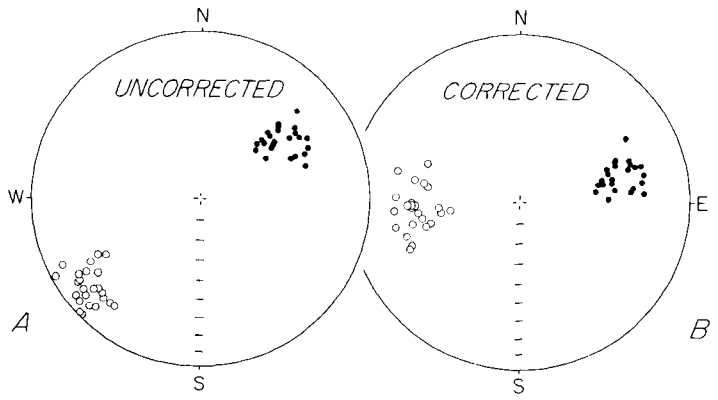


Figure 5. Equal-area plot of palaeomagnetic directions for the Sespe Formation (A) before and (B) after a bedding correction following thermal demagnetization at 400 °C. Note in (B) the approximate 80° clockwise rotation displayed by the declination.

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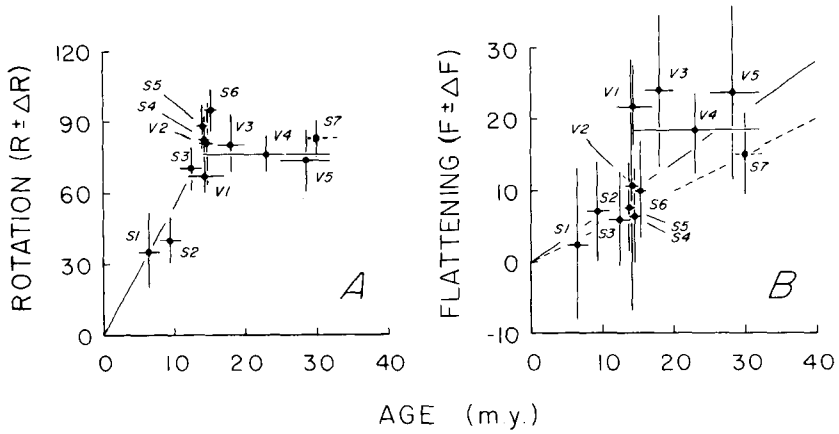


Figure 6. Plot of (A) rotation (difference between observed and expected declination) and (B) flattening (difference between observed and expected inclination) versus age using the methods of Beck (1980) and Demarest (1983) for the Western Transverse Block (Table 2); error bands represent 95 per cent confidence limits. In (B) the solid line is the best-fit line for all the data, and the dashed line is the best-fit line for only sediments (S1–S6: Monterey Formation, Hornafius 1985; S7: Sespe Formation of this report).

1983) and, as previously noted, clockwise rotation of the block in response to dextral shear between the North American and Pacific plates since the early to middle Miocene. The origin of the sediments in the Palaeogene forearc basin that comprise the block (Hamilton 1978; Crouch 1979) depends on which of the hypotheses is correct. For the latter two, a counterclockwise rotation places the deposition on the continent side of the basin, whereas the opposite applies if the rotation were clockwise. The palaeomagnetic record in the Sespe Formation shows that the rotation has been clockwise since the Oligocene. Furthermore, a synthesis of palaeomagnetic data for the Western Transverse Block for the past 30 Myr (Fig. 6, Table 2) shows that the region between the Santa Ynez Fault and Dume Fault acted as a coherent block throughout this period.

CONCLUSIONS

(1) Oxidized claystone and siltstone in the Sespe Formation in Santa Barbara County record a mean declination of about 80° and a mean inclination of about 39°;

Table 2. Palaeomagnetic data for the Western Transverse Block of southern California.

ID	AGE	N	INCL	DECL	ΔI	ΔD	k	α_{95}	R	ΔR	F	ΔF	PLAT	PLONG	REF
S1	5-8	4	52.4	31.3	9.5	15.6	31.3	9.5	34.8	16.8	2.4	10.7	63.8	323.8	A
S2	8-11	9	47.7	35.9	4.9	7.3	109.7	4.9	39.4	9.5	7.1	7.0	59.0	330.5	A
S3	11-14	7	48.8	66.9	4.8	7.3	159.8	4.8	70.4	9.5	6.0	6.9	34.2	314.8	A
V1	12-17	10	32.4	64.0	5.3	6.3	85.5	5.3	67.5	6.9	21.9	5.8	31.0	329.6	B
V2	13.6-15.6	6	42.7	77.5	22.0	30.9	10.2	22.0	80.9	31.4	10.8	22.8	23.4	316.2	C
V3	14-32	52	36.1	72.6	5.1	6.3	15.8	5.1	76.1	7.0	18.2	5.7	25.2	322.4	B
S4	14.2-14.8	14	48.4	79.6	4.6	6.9	75.6	4.6	83.1	9.3	6.4	6.8	24.1	309.7	A
S5	14.2-14.8	26	46.9	85.3	3.6	5.3	64.4	3.6	88.8	8.1	7.9	6.2	19.0	308.3	A
S6	14.8-15.8	12	44.8	91.7	4.9	6.9	80.7	4.9	95.2	9.2	10.0	7.0	13.3	306.8	A
V4	16-20	14	30.3	76.9	12.5	14.5	11.2	12.5	80.3	11.7	24.0	10.7	19.7	323.4	B
V5	25-32	10	30.7	69.6	14.6	17.0	12.0	14.6	74.3	14.3	23.8	12.2	25.9	326.6	B
S7	30	8	38.9	80.1	5.4	6.9	170.0	5.4	84.8	6.8	15.1	4.8	20.1	316.9	D

ID: Locality -- S1, S2, S3, S4, S5, S6: Monterey Fm.; S7: Sespe Fm., V1: Anacapa Is.; V2: Pt. Dume; V3: Sta. Monica Mt.; V4: Sta. Cruz Is. North; V5: San Miguel Is.
AGE: Age in million years
N: Number of sites or units sampled
INCL: Mean inclination (degrees) in stratigraphic coordinates
DECL: Mean declination (degrees) in stratigraphic coordinated
ΔI: α_{95}
ΔD: $\sin^{-1}(\sin\alpha_{95}/\cos I)$
k: Fisher (1953) precision parameter
α_{95}: Alpha-95: radius (degrees) of circle of 95-percent confidence about the mean palaeomagnetic direction
R: Rotation calculated from the equations of Beck (1980) relative to the North American pole for the appropriate time interval
ΔR: Adjusted uncertainty on R from Beck (1980) using the conversion factor from Demarest (1983) where applicable
F: Flattening of inclination
ΔF: Adjusted uncertainty on F from Beck (1980) using the conversion factor from Demarest (1983) where applicable
PLAT: North latitude of the Virtual Geomagnetic Pole (VGP)
PLONG: East longitude of the VGP
REF: References: A -- Hornafius, 1985; B -- Kamerling & Luyendyk, 1985; C -- Liddicoat, 1988; D -- this report

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the inclination is shallow (flat) by about 15° when compared to the expected inclination for this region in the Oligocene from the synthesis of Irving & Irving (1982). The remanent magnetization is believed to be chemical in origin, and it was acquired prior to the most recent folding.

(2) When the mean palaeomagnetic directions in the Sespe Formation are combined with published data cited as reliable for the Western Transverse Block [results for Eocene sandstone on Santa Rosa island are excluded because they might reflect a remagnetization by a nearby intrusive (Kamerling & Luyendyk 1985)], there is evidence that the block was at the latitude of Baja California about 30 Myr ago. Moving northward at a rate of about 0.3° of latitude per million years, the block began to rotate clockwise about 15 Myr ago. That rotation has been episodic into the Pleistocene.

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